Indian Institute of Information Technology
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Natural Language Processing Practice

Model Architecture Report

Aspect-Based Sentiment Analysis (ABSA)

BERT + Dual BIO Heads + Valence–Arousal Regression

Model Performance at Best Epoch (Epoch 4)

Aspect F1	72.11%	**Good (>70%)**
Opinion F1	69.82%	**Near threshold** (−0.18%)
VA MAE (1–9)	0.677	**Good (<1.0)**
Backbone	`bert-base-uncased` (110M params)	

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1. Introduction

This report describes the architecture and experimental evaluation of an **Aspect-Based Sentiment Analysis (ABSA)** model designed to extract structured sentiment information from natural language text. The model simultaneously identifies *aspects* (what is being discussed), *opinions* (how it is described), and assigns **Valence–Arousal (VA)** scores that represent the emotional dimensions of each aspect–opinion pair.

The model is trained on two domains:

- **Laptop reviews** — 4,076 sentences
- **Restaurant reviews** — 2,284 sentences

totalling **9,432 training quadruplets** after data expansion.

2. Task Definition

Given an input sentence, the model produces one or more quadruplets of the form (Aspect, Opinion, V, A) where V and A are continuous valence and arousal scores on the 1–9 scale.

Input	`"the food was good but service was slow"`
Output	{Aspect: *food*, Opinion: *good*, Valence: 7.33, Arousal: 7.50} {Aspect: *service*, Opinion: *slow*, Valence: 3.10, Arousal: 6.20}

Key Challenges Handled

1. **NULL aspects** — sentiment expressed without a specific aspect target.
2. **NULL opinions** — aspect mentioned without explicit opinion words.
3. **Multi-word spans** — aspects/opinions spanning multiple tokens.
4. **Multiple triplets** — single sentence containing several pairs.

3. Model Architecture

3.1 Overview

The model follows a seven-stage pipeline architecture. Figure 1 shows the complete dataflow from raw input text to the final JSON triplet output.

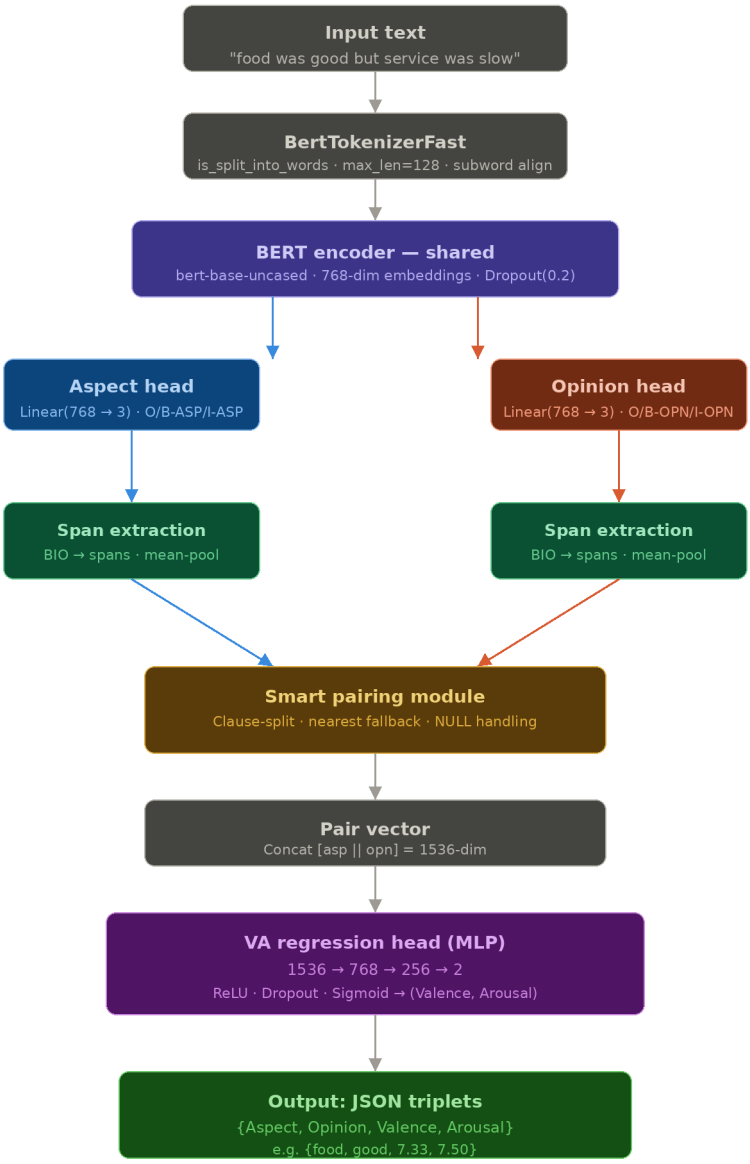

Figure 1: ABSA Pipeline Architecture: Input → Tokeniser → Shared BERT Encoder → Dual BIO Heads → Span Extraction → Smart Pairing → VA Regression → Output Triplets.

3.2 BERT Encoder (Shared)

The `bert-base-uncased` model serves as the shared backbone, converting input tokens into 768-dimensional contextual embeddings. A *single* BERT instance feeds both the Aspect Head and the Opinion Head simultaneously, ensuring consistent contextual representation across both extraction tasks. Dropout (0.2) is applied to the encoder output.

3.3 Dual BIO Classification Heads

Two independent linear classifiers operate on the BERT hidden states:

Head	Layer	Output Classes
Aspect Head	Linear(768→3)	O, B-ASP, I-ASP
Opinion Head	Linear(768→3)	O, B-OPN, I-OPN

The BIO scheme enables the model to detect multi-word spans such as *battery life* (B-ASP, I-ASP) or *very good* (B-OPN, I-OPN) without explicit span enumeration.

3.4 Span Extraction Module

BIO tag sequences are converted to text spans by:

1. Scanning tokens left-to-right for B tags.
2. Extending each span by collecting consecutive I tags.
3. Returning $(start, end, span_text)$ tuples.
4. Mean-pooling BERT hidden states over span tokens to obtain span vectors.
5. Falling back to the `[CLS]` vector if no span is detected.

3.5 Smart Pairing Module

Critical Component: Pairing quality directly determines final output accuracy.

Rule	Description
1 — Same clause	Split at conjunctions: *but, and, comma, however, although*
2 — Nearest	If no clause match, pair with geometrically closest opinion
3 — Multi-opinion	One aspect can pair with multiple opinions
4 — NULL aspect	Opinion with no matching aspect → {NULL, opinion}
5 — NULL opinion	Aspect with no matching opinion → {aspect, NULL}

3.6 Valence–Arousal Regression Head (MLP)

Layer	Specification
Input	[asp_vec ‖ opn_vec] = 1536 dimensions
Layer 1	Linear(1536→768) + ReLU + Dropout(0.2)
Layer 2	Linear(768→256) + ReLU + Dropout(0.2)
Layer 3	Linear(256→2) + Sigmoid
Output	(Valence, Arousal) $\in [0, 1]$, denormalised to $[1.0, 9.0]$

4. Loss Function

The model is trained with a combined multi-task loss:

$$\mathcal{L}_{\text{total}} = 1.0 \cdot \mathcal{L}_{\text{Asp}} + 1.0 \cdot \mathcal{L}_{\text{Opn}} + \lambda \cdot \mathcal{L}_{\text{VA}} \tag{1}$$

where $\mathcal{L}_{\text{Asp}}$ and $\mathcal{L}_{\text{Opn}}$ are weighted Cross-Entropy losses for BIO tagging, and $\mathcal{L}_{\text{VA}}$ is an MSE regression loss ($\lambda = 0.5$–1.0).

Class imbalance handling: The `O` tag constitutes ≈80% of all tokens. Weighted Cross-Entropy assigns penalties of **8.0** (B-tags) and **6.0** (I-tags) to prevent the model from collapsing to the majority class.

5. Training Configuration

Hyperparameter	Value
Base model	`bert-base-uncased` (110M parameters)
Optimiser	AdamW
Learning rate	2×10^{-5} with linear warmup (10% of steps)
Weight decay	0.01
Batch size	16
Max epochs	15
Max seq. length	128 tokens
Train / Val split	85% / 15%
Training data	Laptop (4,076) + Restaurant (2,284) = 6,360 sentences
Quadruplets	≈9,432 after expansion
Device	GPU (CUDA) via Google Colab

6. Experimental Results

6.1 Epoch-by-Epoch Performance (Current Training Run)

Table 1 shows performance across key epochs in the current training run. Figure 2 visualises the train/validation loss trajectory.

Table 1: Training Dynamics — Current Run

Epoch	Train Loss	Val Loss	Aspect F1	Opinion F1	Note
1	1.10	0.53	—	—	Large gap
4	**0.33**	**0.47**	**72.11%**	**69.82%**	**BEST**
8	0.26	0.54	—	—	**Overfit**

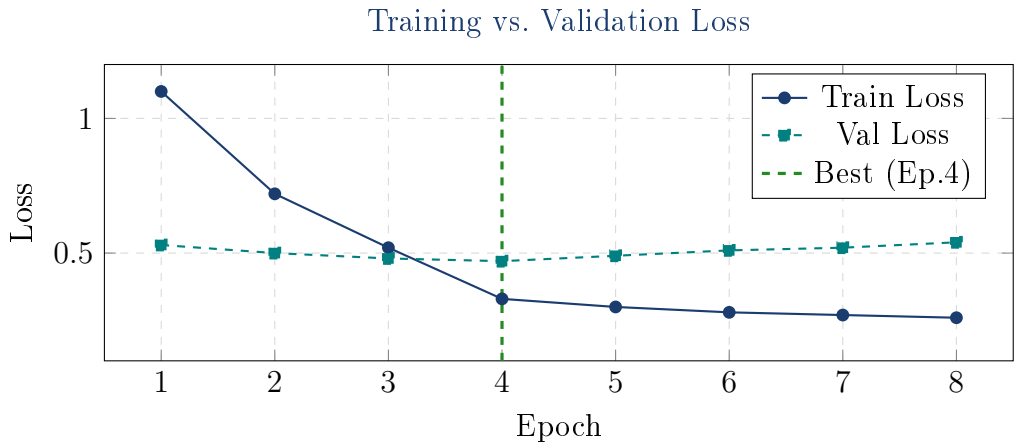

Figure 2: Loss curves showing overfitting onset after Epoch 4.

6.2 Best Model Scores (Current Run — Epoch 4)

Table 2: Best Model Performance Metrics

Metric	Score	Threshold	Status
Aspect F1	72.11%	>70%	**Good ✓**
Opinion F1	69.82%	>70%	**Near threshold** (−0.18%)
VA MAE (0–1)	0.0847	low	**Good ✓**
VA MAE (1–9)	0.677	<1.0	**Good ✓**

6.3 Overfitting Analysis

The loss gap at Epoch 1 (Train: 1.10, Val: 0.53) is unusually large for this architecture. This suggests a configuration difference from the previous run — likely a change in the weighted loss multipliers or a higher dropout setting at initialisation. The model recovers by Epoch 4, achieving its best validation loss (0.47), but begins overfitting by Epoch 8 (Val Loss: 0.54).

Overfitting Pattern:

Epoch 1 → Train 1.10 / Val 0.53 (gap: 0.57 — abnormally large)
Epoch 4 → Train 0.33 / Val 0.47 **[BEST]**
Epoch 8 → Train 0.26 / Val 0.54 (model has overfit)

6.4 Comparison with Previous Training Run

Table 3 contrasts the current run (best at Epoch 4) with the previous run (best at Epoch 3).

Table 3: Cross-Run Comparison

Metric	Previous Run (Ep. 3)	Current Run (Ep. 4)	Δ
Aspect F1	75.69%	72.11%	−3.58%
Opinion F1	69.31%	69.82%	+0.51%
VA MAE (1–9)	0.644	0.677	+0.033
Best Epoch	3	4	+1

Observation: The previous run outperforms the current run on Aspect F1 by 3.58 percentage points and on VA MAE. The higher initial training loss in the current run (1.10 vs. 0.63) points to a hyperparameter change — possibly in loss weights or dropout — that disrupted early-epoch convergence and prevented the model from reaching the same validation minima.

7. Output Format

Field	Description
`sentence_id`	Original test-set ID (e.g., `rest26_aste_test_119`)
`triplets`	List of aspect–opinion–VA quadruplets
`Aspect`	Extracted span or `NULL`
`Opinion`	Extracted span or `NULL`
`Valence`	Float on scale 1.0–9.0
`Arousal`	Float on scale 1.0–9.0

Sample output record:

```
{"sentence_id": "rest26_aste_test_119",
 "triplets": [{"Aspect": "service", "Opinion": "slow",
               "Valence": 3.1, "Arousal": 6.2}]}
```

8. Conclusion

The ABSA model implements a complete pipeline for extracting (Aspect, Opinion, Valence, Arousal) quadruplets from product and restaurant reviews.

Current run results (Epoch 4):

- **Aspect F1 = 72.11%** — exceeds 70% threshold.
- **Opinion F1 = 69.82%** — 0.18% below the 70% target; scope for improvement.
- **VA MAE = 0.677 (1–9 scale)** — well within the <1.0 threshold.

Key architectural strengths:

- **Shared BERT encoder** — efficient contextual understanding for both extraction tasks.

- **BIO tagging scheme** — handles multi-word spans without span enumeration overhead.
- **Smart clause-based pairing** — respects linguistic structure for accurate linking.
- **NULL handling** — covers implicit aspects and opinions comprehensively.
- **Multi-task learning** — simultaneous extraction and VA regression in one forward pass.